

DECORAH, IA, USA

WMO: 725476

Lat: 43.275N

Lon: 91.739W

Elev: 353

StdP: 97.16

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04916

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.5	-20.0	-27.3	0.3	-22.0	-24.8	0.4	-20.0	9.9	-9.9	8.9	-8.1	2.2	310	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	32.1	24.0	30.1	23.1	28.7	22.4	25.4	29.7	24.3	28.6	23.5	27.4	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.7	27.9	22.8	18.3	27.0	22.3	17.7	26.6	80.0	29.8	75.4	28.7	71.9	27.5	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.7	-26.6	33.7	3.5	1.9	-29.2	35.1	-31.2	36.2	-33.2	37.3	-35.8	38.7		
(5)			-26.7	27.0	3.1	1.2	-28.9	27.9	-30.7	28.6	-32.4	29.3	-34.6	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-6.9	-5.3	1.7	9.0	15.3	20.6	22.6	20.8	17.2	10.4	2.9	-4.3
	DBStd	11.70	7.07	6.97	6.89	5.65	4.85	3.66	3.13	2.95	4.47	5.19	6.00	6.35
	HDD10.0	2048	524	428	271	84	9	0	0	0	3	60	223	444
	HDD18.3	3885	783	661	517	283	121	18	4	10	73	251	463	702
	CDD10.0	1584	0	1	14	53	172	317	392	334	219	71	11	1
	CDD18.3	381	0	0	2	3	25	85	137	86	39	4	0	0
	CDH23.3	3100	0	0	9	37	232	713	1167	603	301	37	1	0
	CDH26.7	863	0	0	1	8	66	213	364	136	69	6	0	0
Wind	WSAvg	2.8	3.1	3.1	3.4	3.7	3.1	2.5	1.9	1.9	2.3	2.8	3.1	3.0
Precipitation	PrecAvg	936	27	25	47	101	114	147	125	123	80	61	53	33
	PrecMax	1196	82	61	101	249	312	262	274	374	166	165	156	77
	PrecMin	624	7	1	3	44	44	69	26	26	32	17	3	7
	PrecStd	143	15	16	28	50	54	64	60	73	42	38	37	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	13.0	24.0	27.4	32.1	33.0	34.0	32.1	31.3	27.5	21.3	12.9
		MCWB	5.9	9.3	17.5	17.2	22.3	24.2	25.0	24.5	22.4	17.6	14.2	11.4
	2%	DB	6.1	9.0	18.7	23.0	27.9	31.2	32.2	29.9	28.7	23.8	17.4	7.7
		MCWB	3.5	7.0	15.0	14.6	19.1	23.3	24.8	23.4	21.8	16.4	12.5	5.7
	5%	DB	3.0	6.2	15.0	21.0	26.0	29.0	30.2	27.8	27.1	21.2	14.0	5.3
		MCWB	1.4	3.8	11.0	13.5	18.1	22.0	23.8	22.4	20.4	15.5	10.1	3.3
	10%	DB	2.1	3.7	11.1	17.8	23.0	27.5	28.7	27.0	24.9	18.7	12.0	3.0
		MCWB	1.0	1.7	8.0	11.7	16.8	20.9	23.0	21.7	19.1	14.5	8.5	1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	10.2	18.0	18.8	23.9	26.1	27.6	25.9	24.6	20.9	15.6	12.1
		MCDB	9.7	12.8	23.0	24.7	28.5	31.0	32.0	29.7	28.6	23.2	19.6	13.4
	2%	WB	3.5	7.1	15.3	16.4	21.6	24.6	25.7	24.5	23.0	18.4	13.3	6.3
		MCDB	5.7	9.0	17.8	21.4	26.1	29.2	29.8	28.1	27.3	20.8	16.4	7.5
	5%	WB	2.0	4.0	11.3	14.6	19.8	23.4	24.7	23.5	21.8	16.3	10.5	3.8
		MCDB	3.4	6.1	14.9	19.3	24.0	27.8	29.1	26.9	25.3	20.1	13.6	5.4
	10%	WB	0.7	1.9	7.8	12.5	18.0	22.0	23.7	22.5	20.3	14.5	8.2	2.1
		MCDB	1.7	3.5	10.5	17.2	21.9	26.7	28.0	25.8	23.4	18.7	10.9	3.3
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.1	8.5	10.1	10.2	9.8	9.8	9.8	10.9	9.9	8.2	7.1
		MCDBR	8.6	9.0	12.5	14.7	13.0	11.2	10.4	10.5	12.4	13.0	11.7	8.5
		MCWBR	6.8	7.3	8.0	8.1	6.7	6.3	5.6	5.9	6.7	7.4	8.1	7.1
	5% WB	MCDBR	8.0	9.0	11.9	12.5	11.1	10.3	9.4	9.4	9.9	10.8	10.8	8.2
		MCWBR	6.7	7.3	8.2	8.2	7.0	6.8	5.9	6.1	6.1	8.0	8.4	7.2
Clear-Sky Solar Irradiance	taub	0.267	0.291	0.319	0.365	0.394	0.407	0.423	0.404	0.374	0.327	0.291	0.269	
	taud	2.391	2.331	2.391	2.310	2.287	2.285	2.239	2.303	2.360	2.486	2.506	2.435	
	Ebn at Noon	878	908	924	900	879	865	846	851	851	859	846	842	
	Edh at Noon	76	97	104	123	130	130	135	123	107	83	69	67	
All-Sky Solar Radiation	RadAvg	1.79	2.82	3.80	4.57	5.42	6.02	6.18	5.29	4.31	2.78	1.85	1.36	
	RadStd	0.18	0.29	0.42	0.42	0.40	0.47	0.32	0.36	0.33	0.38	0.15	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page